



Bullwhip Effect

DYNAMIC AMPLIFICATION IN SUPPLY CHAINS

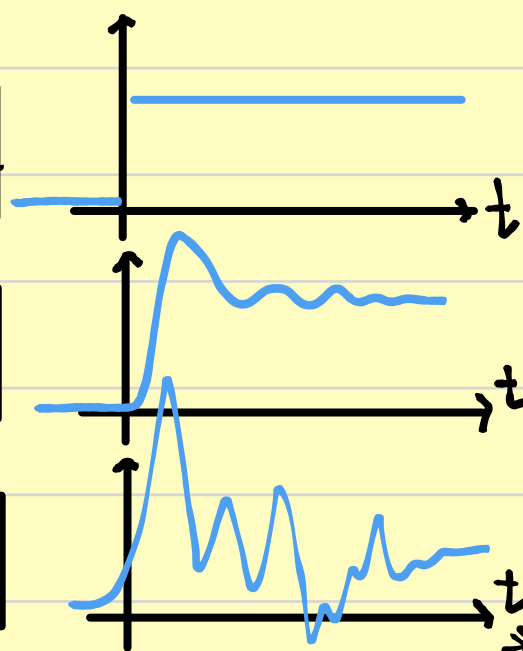
Bullwhip: demand variability increases as we move upstream in supply chains.

Example:

Customer demand

Retailer order

Wholesale order



NOTATION:

$d(t) \equiv$ customer demand rate

$o(t) \equiv$ order rate placed

$a(t) \equiv$ arrival/replenishment rate

$I(t) \equiv$ Inventory deviation from target

$K \equiv$ Inventory correction gain

$\tau \equiv$ replenishment/lead time lag

$u(t) \equiv$ unit/step function

* Remark: we work with deviations around operating point.

$d(t)=0$ does not mean no demand. It means it is at its normal planned level!

Typical causes:

- demand forecasting
- replenishment lead time
- order batching
- inventory corrections (play safe)
- overreaction shortages

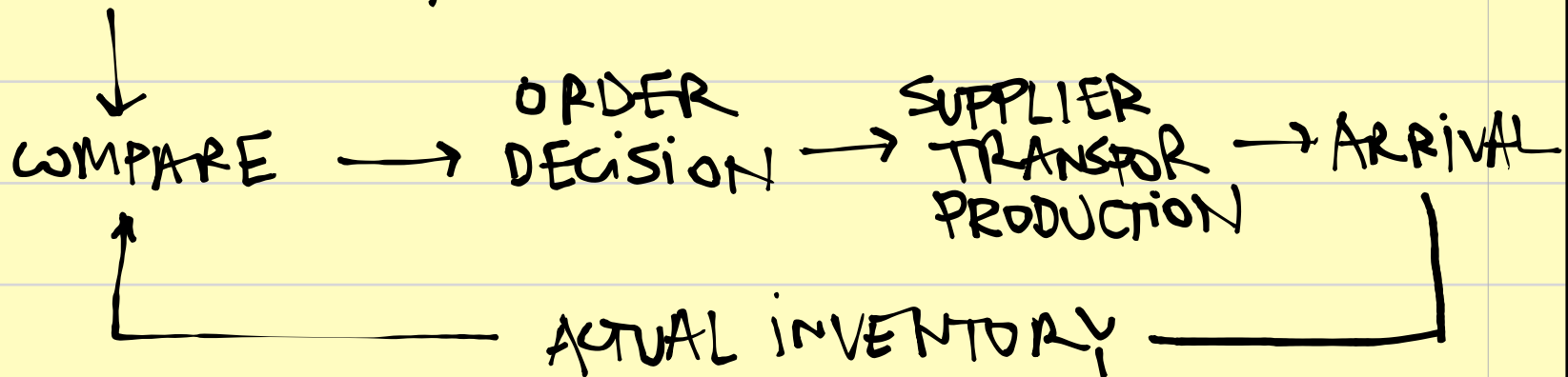
Mathematically these can be summarized by two ingredients

1) FEEDBACK CORRECTION

2) DELAY / LAG

The basic replenishment feedback :

TARGET INVENTORY



A simple stock-flow model for one supply-chain stage

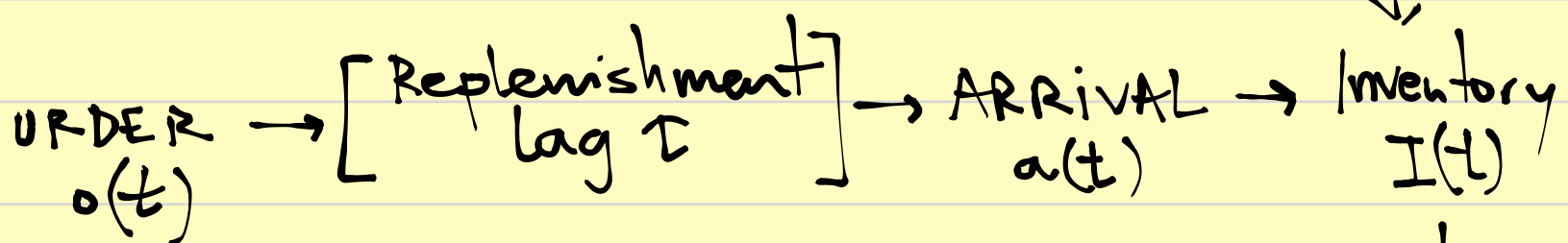
• Inventory balance : $\dot{I}(t) = a(t) - d(t)$

This is a stock conservation equation.

• Orders do not arrive instantly: we model the arrival rate as a first-order lag of order rate.

$$\tau \dot{a}(t) + a(t) = o(t)$$

customer demand
 $d(t)$



• Ordering rule: order enough to replace demand & correct inventory error.

$$o(t) = d(t) - k I(t)$$

*The - sign comes from the fact that if Inventory is below target, then $I(t) < 0$, so $-k I(t) > 0$ and the order is increased.

The full model:

Feedback model for Bullwhip.

$$\dot{I}(t) = a(t) - d(t) \quad (1)$$

$$\tau \dot{a}(t) + a(t) = o(t) \quad (2)$$

$$o(t) = d(t) - k I(t) \quad (3)$$

We want to answer **How STRONGLY DOES THE ORDER $o(t)$ REACT TO DEMAND $d(t)$?**

We want the transfer function $\frac{O(s)}{D(s)}$

• Apply Laplace to (1): $sI(s) = A(s) - D(s) \quad (4)$

• Apply Laplace to (2): $\tau s A(s) + A(s) = O(s)$
 $\rightarrow A(s) = \frac{O(s)}{\tau s + 1} \quad (5)$

• Apply Laplace to (3): $O(s) = D(s) - k I(s) \quad (6)$

• From (4): $I(s) = \frac{A(s) - D(s)}{s}$

• From (5): $I(s) = \frac{\frac{O(s)}{\tau s + 1} - D(s)}{s}$

• From (6): $O(s) = D(s) - k \cdot \left[\frac{\frac{O(s)}{\tau s + 1} - D(s)}{s} \right]$

$$\rightarrow sO(s) = sD(s) - k \frac{O(s)}{\tau s + 1} + k D(s)$$

$$\rightarrow \left[s + \frac{k}{\tau s + 1} \right] O(s) = (s + k) \cdot D(s) \rightarrow$$

$$\rightarrow \boxed{\frac{O(s)}{D(s)} = \frac{(s+k)(\tau s+1)}{\tau s^2 + s + k}}$$

This eq. describes how customer demand deviation become upstream order deviation.

Poles and damping:

$$\tau s^2 + s + k = 0 \rightarrow s^2 + \frac{s}{\tau} + \frac{k}{\tau} = 0$$

The standard 2nd order form: $s^2 + 2\zeta\omega_n s + \omega_n^2$

In our case:

$$\omega_n^2 = \frac{k}{\tau} \rightarrow \omega_n = \sqrt{\frac{k}{\tau}}$$

$$2\zeta\omega_n = \frac{1}{\tau} \rightarrow \zeta = \frac{1}{2\sqrt{k\tau}} \rightarrow \zeta = \frac{1}{2\sqrt{k\tau}}$$

The damping ratio becomes smaller when k increases or when τ increases.

So, a more aggressive correction and longer lead time make both the system more oscillatory.

Example with: step $u(t)$ input.

$$d(t) = \Delta d u(t)$$

simple case: $\Delta d = 1$ $K = 1$ $\tau = 4$

$$D(s) = \frac{1}{s} \quad \& \quad \frac{O(s)}{D(s)} = \frac{(s+1)(4s+1)}{4s^2+s+1} \rightarrow$$

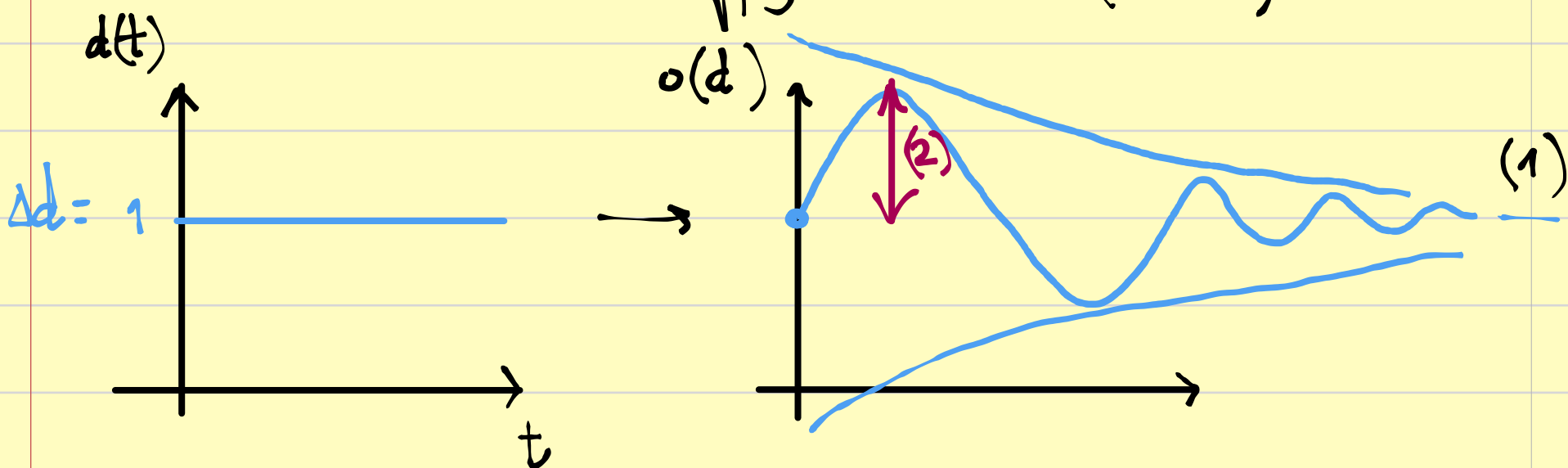
$$\rightarrow O(s) = \frac{(s+1)(4s+1)}{s(4s^2+s+1)} = \frac{4s^2+5s+1}{s(4s^2+s+1)} =$$
$$= \frac{A}{s} + \frac{B}{4s^2+s+1} = \frac{1}{s} + \frac{4}{4s^2+s+1}$$

$$4s^2+5s+1 = A(4s^2+s+1) + Bs$$

$$A=1; B=4$$

$$4s^2+s+1 = 0 \rightarrow s = \frac{-1 \pm \sqrt{1-16}}{8} = \frac{-1 \pm \sqrt{15}j}{8}$$

$$\rightarrow \dots \rightarrow o(t) = 1 + \frac{8}{\sqrt{15}} e^{-t/8} \sin\left(\frac{\sqrt{15}}{8} t\right)$$



The system is stable. (1)

A stable system can still show a whip if it overshoots. (2)

Causes and countermeasures in control language.

The BE is caused by certain decisions and structures.

- gain / delay / filtering / batching / ...

CAUSE 1. Long Lead Time

- Lead time appears as τ , approximated by $\frac{1}{\tau s + 1}$
- larger τ means slower replenishment.
- From $\xi = \frac{1}{2\sqrt{k\tau}}$, larger τ means smaller damping.
- Larger Lead-time τ means more oscillation risk

LITTLE' LAW: $LT = \frac{WIP}{TH}$

$\left[\frac{\text{Parts}}{\text{Time}} \right]$

/

$\left[\frac{\text{Parts}}{\text{Time}} \right]$

CAUSE 2. AGGRESSIVE INVENTORY CORRECTION

- Inventory correction gain: k
- Larger k means react strongly to inventory error
- From $\xi = \frac{1}{2\sqrt{k\tau}}$, larger k also reduces damping.
- Aggressive correction is a trade-off: can reduce

inventory faster, but increase oscillations.

Why should logistic engineers care?

BE causes:

- 1) high inventory
- 2) low service reliability
- 3) production instability
- 4) transport peaks
- 5) warehouse overload
- 6) poor capacity planning
- 7) increases cost

Policies are:

- A) base stock control
- B) KANBAN/CONWIP
- C) order smoothing
- D) reduce lead time

